

Classification Trees

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Motivation

Classification trees are a supervised learning method for predicting categorical outcomes.

- Interpretable models
- Nonlinear decision boundaries
- Handle mixed data types

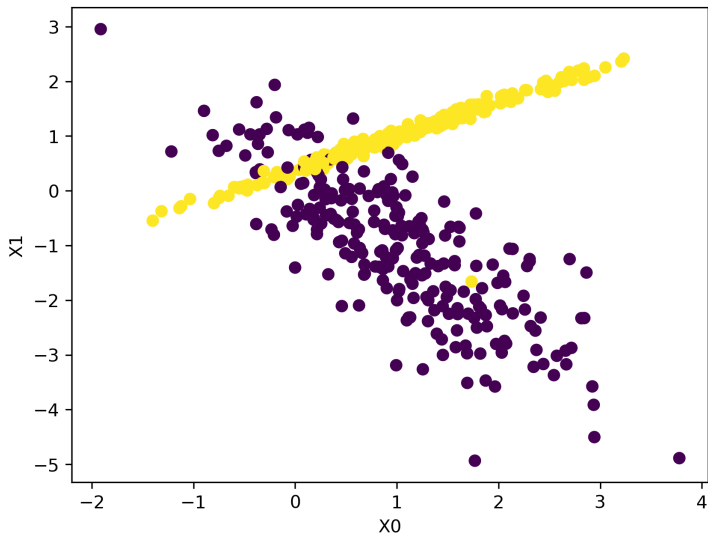
Supervised Learning Setup

We observe:

- Features $X \in \mathbb{R}^{n \times p}$
- Labels $y \in \{1, \dots, K\}$

Goal: learn a function $f(X)$ that predicts y .

Example Data



What is a Classification Tree?

A recursive partitioning of feature space into regions.

- Internal nodes: decision rules
- Leaves: class predictions

Binary Splits

At each node, choose:

- Feature j
- Threshold t

$X_j \leq t$ vs $X_j > t$

Impurity Measures

Gini: $G = 1 - \sum_k p_k^2$

Entropy: $H = - \sum_k p_k \log p_k$

Evaluating a Candidate Split

Parent: $p_1 = 0.5, p_2 = 0.5$

$$G_{parent} = 0.5$$

After split:

- Left: $G = 0.0$
- Right: $G = 0.32$

Weighted impurity decreases $\rightarrow$ good split.

Worked Example: Gini Reduction

Parent node:

$$p_1 = 0.5, p_2 = 0.5$$

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After split:

Left node (pure):

$$G_L = 1 - (1^2 + 0^2) = 0$$

Right node (mixed):

$$p = (0.8, 0.2)$$

$$G_R = 1 - (0.8^2 + 0.2^2) = 1 - (0.64 + 0.04) = 0.32$$

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Weighted impurity:

$$G_{split} = 0.4(0) + 0.6(0.32) = 0.192$$

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Weighted impurity:

$$G_{split} = 0.4(0) + 0.6(0.32) = 0.192$$

Impurity reduction:

$$\Delta G = 0.5 - 0.192 = 0.308$$

Recursive Partitioning

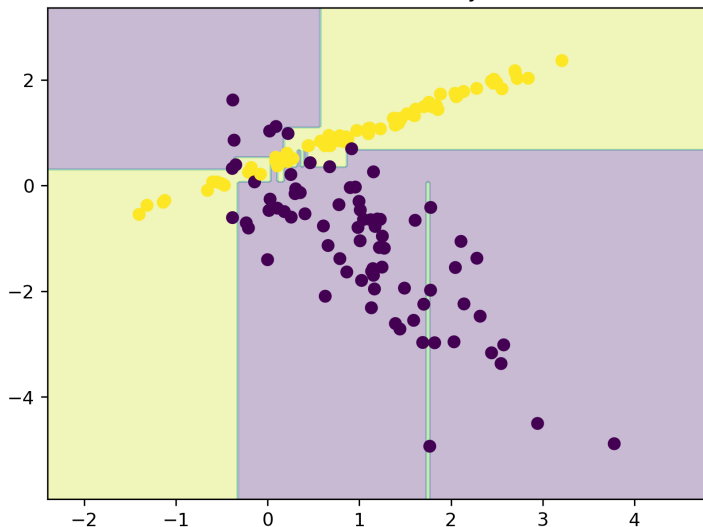
- 1 Find best split (Maximize ΔG)
- 2 Partition data
- 3 Repeat

Stopping Criteria

- Max depth
- Min samples per leaf
- No improvement

Overfitting

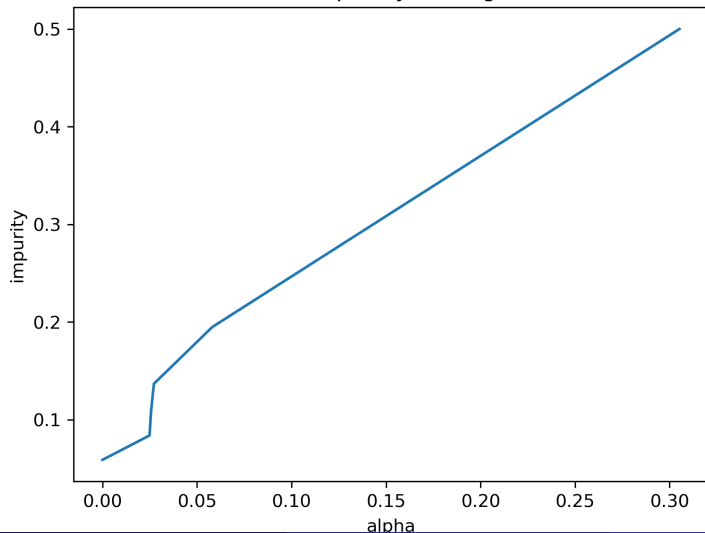
Decision Boundary



Pruning

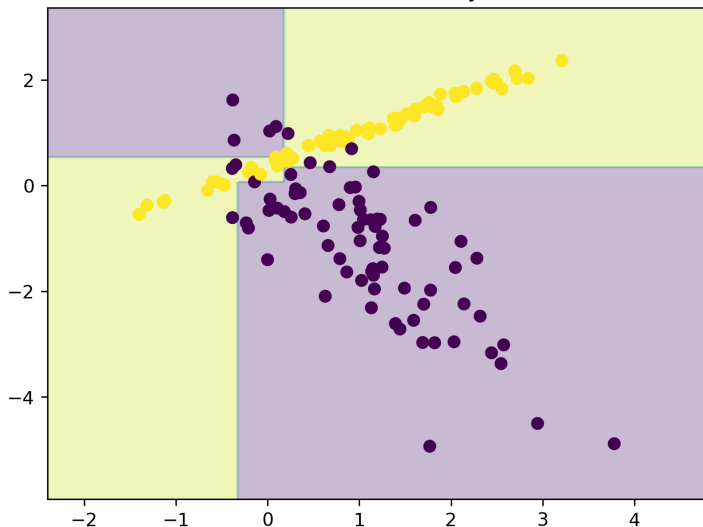
$$R_\alpha(T) = R(T) + \alpha|T|$$

Cost-Complexity Pruning Path

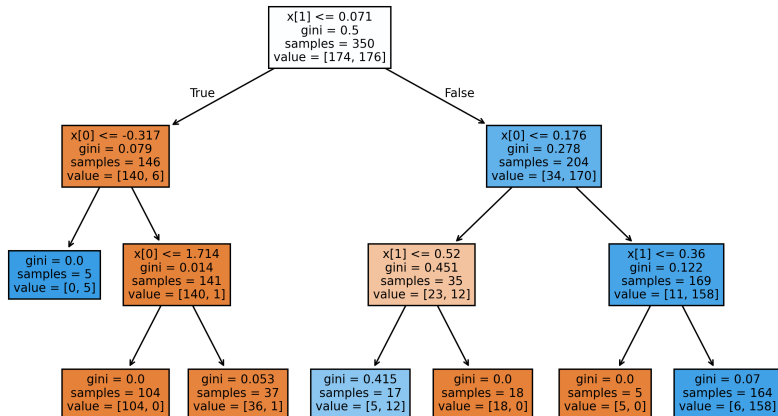


Decision Boundaries

Decision Boundary



Tree Structure



Classifying a New Observation with the Fitted Tree

New observation: $x = (x_0, x_1) = (0.10, 0.20)$

- 1 Root: is $x_1 \leq 0.071$?
 $0.20 \leq 0.071 \Rightarrow$ **False** $\rightarrow$ go right

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- ① Root: is $x_1 \leq 0.071$?
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- ② Next: is $x_0 \leq 0.176$?
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- ③ Next: is $x_1 \leq 0.52$?
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- ④ Leaf reached: value = $[5, 12]$
Majority class: **Class 1**

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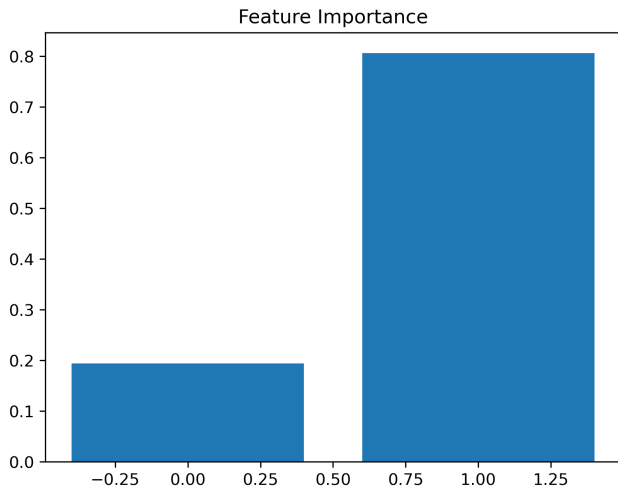
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Prediction: $\hat{y}(x) = 1$

Leaf probabilities:

$$\hat{P}(y = 0 \mid x) = \frac{5}{17}, \quad \hat{P}(y = 1 \mid x) = \frac{12}{17}$$

Feature Importance



Advantages

- Interpretable
- Nonlinear
- No scaling

Limitations

- Unstable
- Overfitting
- Axis-aligned

Spatial Perspective

Attribute regions $\neq$ geographic regions

Extensions

- Random forests
- Gradient boosting

Summary

- Recursive partitioning
- Greedy splits
- Bias–variance tradeoff